



AI Agent Support for Data Analysis

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Motivation

European XFEL (EuXFEL) and the Informatics department at the University of Hamburg are developing an AI agent to support users during **offline data analysis**. This assistant will function like ChatGPT, Claude, or Gemini and will be directly integrated into the EuXFEL tool chain (Jupyter Notebook, Zulip, etc.). It will have access to EuXFEL-specific data analysis knowledge, can retrieve and contextualize documentation for your needs, and assist you in writing analysis code. **Help us shaping this AI agent by completing our survey.**

We need your opinion!

- Have you **previously used EuXFEL’s data analysis tools** (e.g., Python library EXtra, myMDC, DAMNIT)?
- What **positive aspects** of the **offline** data analysis process have you experienced? Please focus on aspects related to **the exchange of knowledge/information** (e.g., support from EuXFEL members, the information provided by the tools, and the tool documentation).
- What **negative aspects** of the **offline** data analysis process have you experienced? Please focus on aspects related to **the exchange of knowledge/information**.
- Imagine you could use a specialized AI tool, which is similar to ChatGPT but provided by EuXFEL. What **tasks should the tool support you** with during offline data analysis?
- Are you a user or a member of EuXFEL?
- How many experiments have you participated in at EuXFEL?



(<https://umfragen.uni-hamburg.de/index.php/459382>)

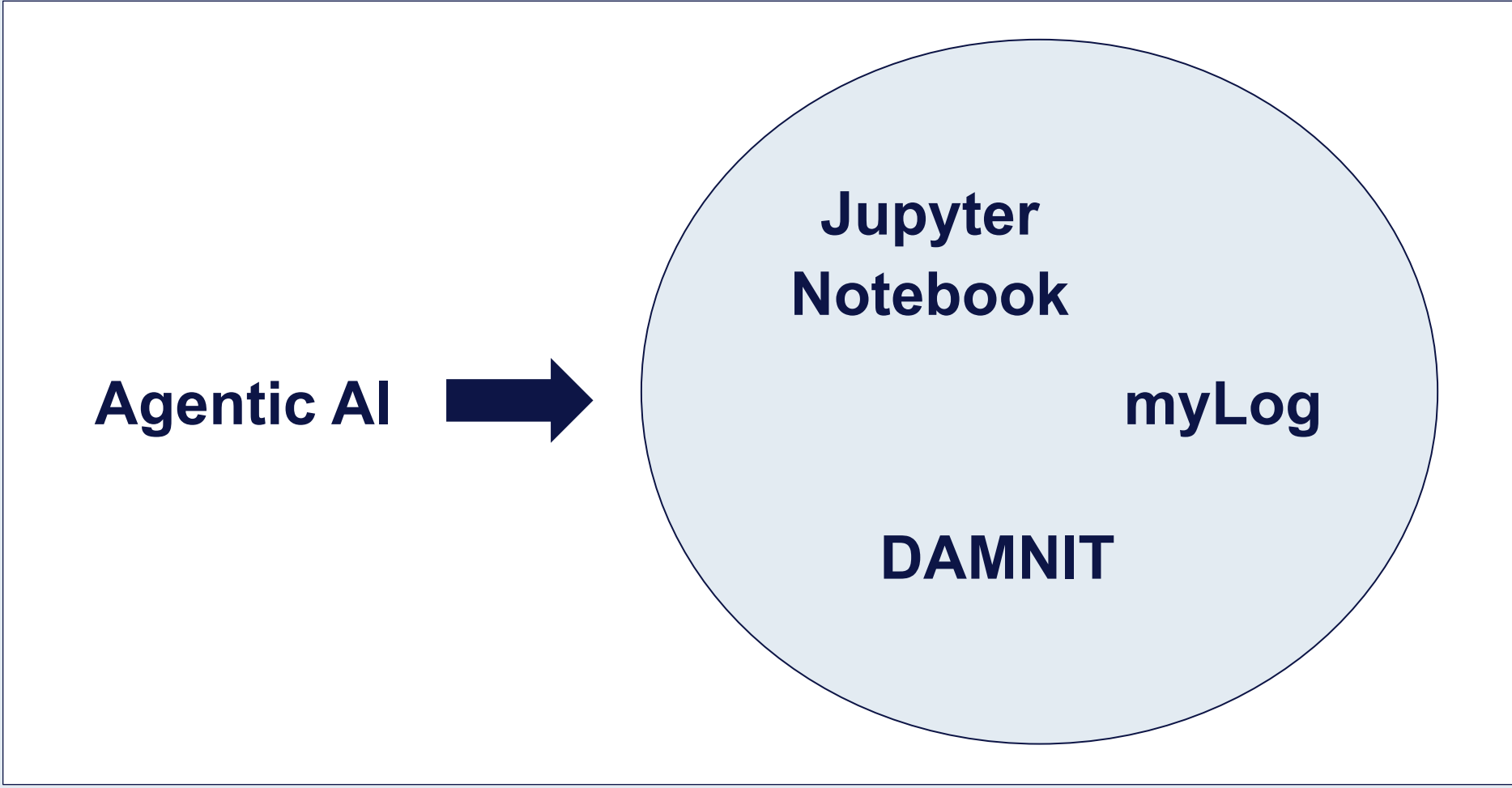
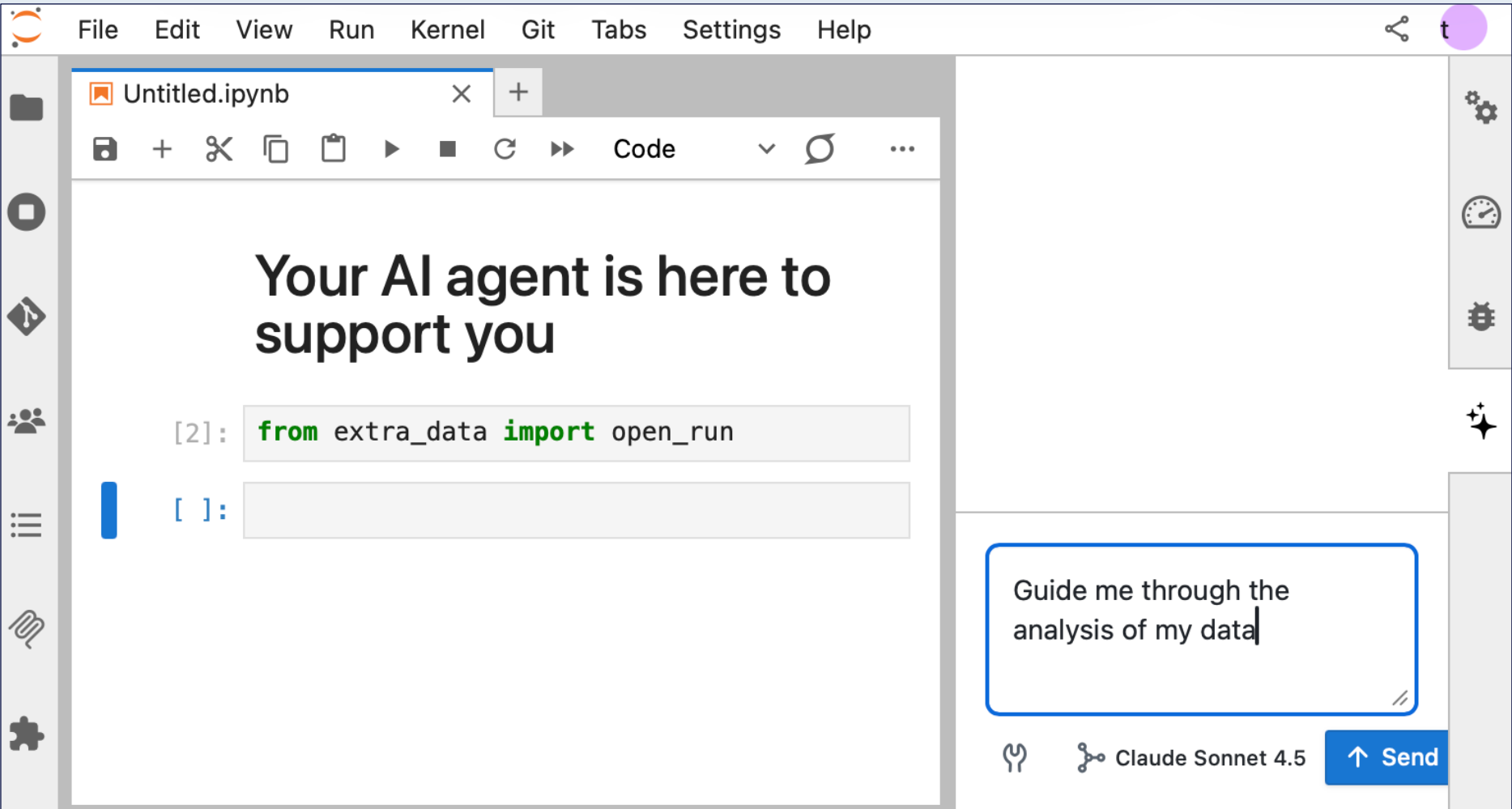
Previous Studies

AI-powered Visual Studio Code extension to generate source code documentation:
H. A. Kruse, T. Puhlfürß and W. Maalej, "Can Developers Prompt? A Controlled Experiment for Code Documentation Generation," 2024 IEEE International Conference on Software Maintenance and Evolution (ICSME), Flagstaff, AZ, USA, 2024, pp. 574-586

AI-powered knowledge base for issue tracker systems and documents:
L. Ciepielowski, T. Puhlfürß and W. Maalej, "An LLM-Based Tool for Knowledge Retrieval from (Heterogeneous) Issue Tracking Systems", master thesis, 2024

Analysis of ethical AI requirements considered in theory and practice:
T. Puhlfürß, J. Butzke and W. Maalej, "Model Cards Revisited: Bridging the Gap Between Theory and Practice for Ethical AI Requirements," 2025 IEEE 33rd International Requirements Engineering Conference (RE), Valencia, Spain, 2025, pp. 280-291

Potential Integration



Acknowledgements

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